

- 6 (a) Two cylindrical resistors M and N of the same material are connected parallel in Fig. 6.1. The mass of M is twice the mass of N and the radius of M is half the radius of N.

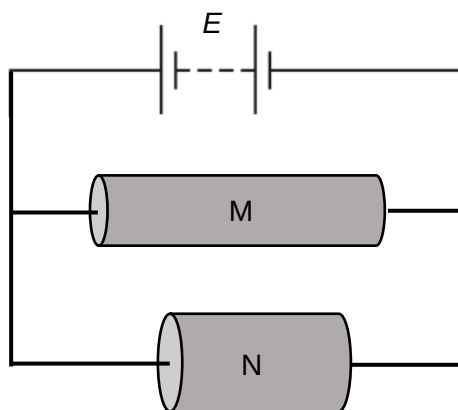


Fig. 6.1 (not to scale)

Determine the ratio

- (i) $\frac{\text{resistance of M}}{\text{resistance of N}}$,

$$\frac{\text{mass of M}}{\text{mass of N}} = \frac{\text{density}_M \text{Volume}_M}{\text{density}_N \text{Volume}_N} = \frac{L_M}{L_N} \times \frac{A_M}{A_N} = \frac{L_M}{L_N} \times \frac{1}{4} = 2, \frac{L_M}{L_N} = 8$$

$$\frac{A_M}{A_N} = \frac{\pi r_M^2}{\pi r_N^2} = \left(\frac{1}{2}\right)^2 = \frac{1}{4}$$

$$\frac{\text{resistance of M}}{\text{resistance of N}} = \frac{\rho_M L_M}{A_M} \times \frac{A_N}{\rho_N L_N} = \frac{L_M}{L_N} \times \frac{A_N}{A_M} = 8 \times 4 = 32$$

ratio = [2]

- (ii) $\frac{\text{average drift speed of electrons in M}}{\text{average drift speed of electrons in N}}$.

$$I = Anvq, \text{ for same charge carriers and same material, } I \propto Av$$

$$\text{for same potential difference, } I \propto \frac{1}{R}, \text{ so, } v \propto \frac{1}{AR}$$

$$\frac{v_M}{v_N} = \frac{A_N R_N}{A_M R_M} = 4 \times \frac{1}{32} = 0.125$$

ratio = [2]

- (b) A cell of electromotive force (e.m.f.) 1.5 V and internal resistance $1.0\ \Omega$ is connected to a resistor X and resistor Y as shown in Fig. 6.2.

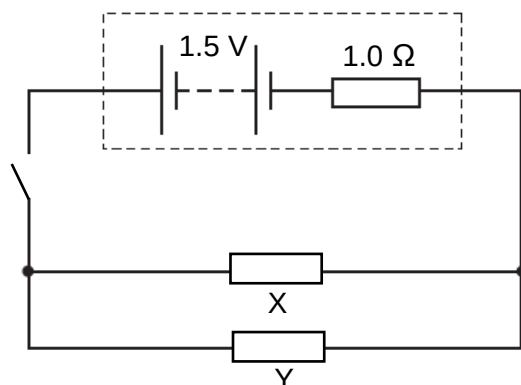


Fig. 6.2

Resistor X has resistance $2.0\ \Omega$ while resistor Y has a resistance of $6.0\ \Omega$.

- (i) Show that the current in the cell is 0.60 A when the switch is closed.

$$\text{Total resistance, } R_T = 1.0 + \frac{1}{\frac{1}{2.0} + \frac{1}{6.0}} = 2.5\ \Omega$$

$$E = I R_T$$

$$I = 1.5 / 2.5 = 0.60\ \text{A. (Shown)}$$

[2]

- (ii) Determine the energy dissipated in the cell when the switch is closed for 8.0 minutes.

$$\text{Energy dissipated} = I^2 r t = (0.60)^2 (1.0)(8.0 \times 60) = 172.8 = 170\ \text{J}$$

$$\text{energy dissipated} = \dots\dots\dots \text{J [1]}$$

- (iii) Resistor Y is replaced with a component Z with similar resistance value. When the temperature increases, the resistance of component Z decreases. State and explain the change to the power dissipated in the cell when temperature increases.

When temperature increased and resistance of Z decreases, the total current in the cell will increase. Thus, the power dissipated in the cell increases.

.....[1]

(iv) The I - V characteristic of X and Z are given in Fig. 6.3.

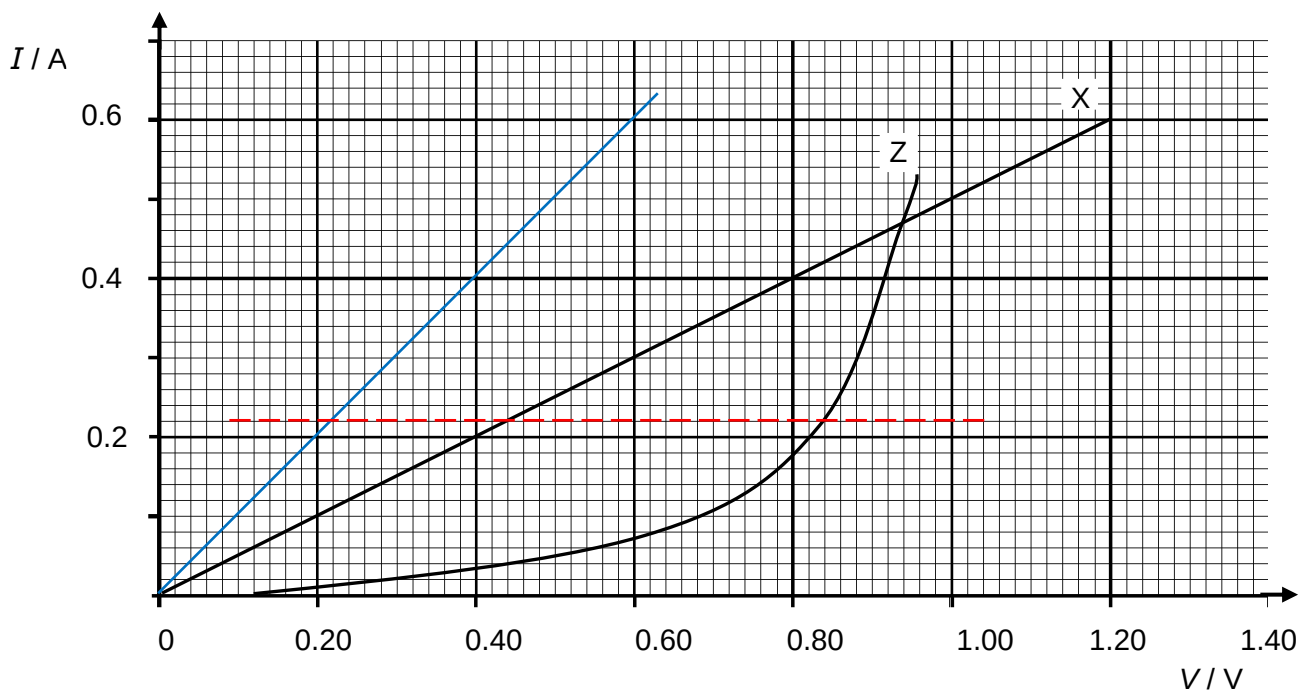


Fig. 6.3

The circuit is now reconnected such that resistor X and component Z are in series with the same cell. Using the Fig. 6.3, or otherwise, determine the potential difference across component Z.

Since the components are connected in series, the current passing through the components are the same, draw a line to find the total p.d. across equal to 1.5 V.

p.d across the X and $1.0\ \Omega$ is 0.66V, and p.d across Z is 0.84V

potential difference = V [2]

[Total: 10]

- 7 A magnet rotates inside a shaped soft iron core. A coil is wrapped around the iron core as shown